

ARMELLE BALLIAN

Born on 18.12.1995 in Saint-Étienne, France

Currently at: Climatology and the Biosphere, Geo- and Environmental Research Center
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<https://scholar.google.com/citations?user=ald4LewAAAAJ&hl=en>

FIELDS OF SPECIALIZATION

Continental paleoclimate and paleoenvironmental dynamics, surface processes, geomorphology, paleoaltimetry, terrestrial carbonates, paleosols, fault gouges, karst systems, cave weather monitoring, stable isotope geochemistry ($\delta^{13}\text{C}$, $\delta^{17}\text{O}$, $\delta^{18}\text{O}$, $\delta^2\text{H}$), carbonate clumped (Δ_{47}) isotope thermometry.

EDUCATION

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|-----------|---|
| 2021–2025 | Ph.D.
Senckenberg Biodiversity and Climate Research Centre, Frankfurt am Main, Germany
Thesis: <i>Miocene paleoclimate and paleoaltimetry reconstructions of the European Alps</i>
Appreciation: magna cum laude
Advisors: Prof. Dr. Andreas Mulch & Ass.-Prof. Dr. Maud Meijers |
| 2017–2020 | Master of Sciences (M.Sc.) in Geology
Albert-Ludwigs-Universität, Institute of Earth and Environmental Sciences, Freiburg im Breisgau, Germany |
| 2016–2017 | Bachelor of Sciences (B.Sc.) in Earth Sciences ERASMUS
Albert-Ludwigs-Universität, Institute of Earth and Environmental Sciences, Freiburg im Breisgau, Germany |
| 2014–2016 | Bachelor of Sciences (B.Sc.) in Earth Sciences
Université Jean Monnet, Saint-Étienne, France |

PROFESSIONAL RECORD

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|--------------|---|
| 2025–Present | Research associate
Department of Climatology and the Biosphere
University of Tübingen, Germany |
| 2020–2021 | Laboratory Technician
Department of Hydrogeochemistry, Laboratoire de Géologie de Lyon
Université Jean Monnet, Saint-Étienne, France |

2018–2020

Student Assistant

Department of Sedimentary Geology and Quaternary Research
Albert-Ludwigs-Universität, Freiburg im Breisgau, Germany

TEACHING EXPERIENCE

Summer Sem. 2026

Lecturer and co-coordinator

Environmental Isotope Chemistry (M208), University of Tübingen
Climate Dynamics and Climate Modeling (M322), University of Tübingen

Lab practicals

Environmental Isotope Chemistry (M208), University of Tübingen

Excursions (one day field trip)

"Excursion to the Swabian Alb - Karst Processes at the Surface"

"Excursion to the Swabian Alb - Karst Processes below the Surface"

Winter Sem. 2025/2026

Lecturer and co-coordinator

Global Change (M229), University of Tübingen

Summer Sem. 2025

Guest seminar lecture and lab practicals

Environmental Isotope Chemistry (M208), University of Tübingen

Excursion (one day field trip)

"Excursion to the Swabian Alb - Karst Processes at the Surface"

STUDENT SUPERVISION

Bachelor thesis

Summer Semester 2026

Jule Christmann (Water stable isotopes and PFAS in the Swabian Alb karst system)

Summer Semester 2025

Judith Baum (Stable isotope compositions in precipitation on the Swabian Alb)

Master thesis

Summer Semester 2026

Desirée LoTriglia (Tracing isotopes from rain to calcite)

GRANTS

September 2025

3000€, **Geogenomic Archaeology Campus Tübingen**

Funded project: "EARTH MEMORY - Caves as Memory Archives of Climate and Life"

HONORS AND AWARDS

September 2024

Student Poster Presentation Award 3rd place

GeoSaxonia 2024, Dresden, Germany

- December 2023 **Outstanding Student Presentation Award (OSPA)**
AGU Fall Meeting 2023, San Francisco, USA
- May 2023 **European Consortium for Ocean Research Drilling (ECORD) scholarship**
for attending the Urbino Summer School in Paleoclimate, Urbino, Italy

INVITED SPEAKER

- June 2026 GFZ Helmholtz Centre for Geosciences, Potsdam, Germany
- October 2023 Centre de Recherche et d'Enseignement en Géosciences de l'Environnement (CEREGE), Aix-en-Provence, France

SERVICE AND PROFESSIONAL MEMBERSHIPS

- 2026–Present Past Global Changes (PAGES) - Member of the "Hydroclimate2k" working group (<https://pastglobalchanges.org/user/10729>)
- 2025–Present Geogenomic Archaeology Campus Tübingen (GACT) - Steering committee member (<https://geogenarch-tuebingen.de/leibniz-sciencecampus/>)
- 2023–2025 Leibniz PhD Network - PhD Representative and Coordinator of the Working Group Communication (<https://leibniz-phd.net/>)
- 2022–2024 Young Scientists - Local Speaker (www.senckenberg.de/en/career/young-scientists/)

REVIEWER

Peer reviewer for *Global and Planetary Change*, ISSN: 0921-8181

CONFERENCE CHAIR

- September 2024 Convener at GeoSaxonia 2024, Dresden, Germany
Session: 'Interactions between mountain building, climate and biodiversity'

PARTICIPATION IN WORKSHOPS AND SUMMER SCHOOLS

- May 2026 **IMPACT Workshop - Past abrupt changes, thresholds, and tipping points: Definitions, Identifications, and Context**
PAGES Impact Working Group, online
- September 2025 **Fit für die Lehre Teil 1 – Hochschuldidaktische Grundlagen**
Hochschuldidaktikzentrum (HDZ) Baden-Württemberg, Tübingen

- November 2024 **Paleontological Field School, Armenia**
organized by the JURASSICA Museum (Porrentruy, Switzerland) and the Institute of the Geological Sciences of the National Academy of Sciences of the Republic of Armenia (Yerevan, Armenia)
- July 2023 **DMG Data Science Short Course**
organized by the University of Frankfurt, Frankfurt am Main, Germany
- July 2023 **Urbino Summer School in Paleoclimate**
organized by ECORD, Urbino, Italy
- June 2021 **'Principles, applications and new frontiers in isotope geochemistry (PAIG)' Summer School**
organized by the Lake Como School, online

PEER-REVIEWED PUBLICATIONS

- [1] **Ballian, A.**, Meijers, M. J. M., Boateng, D., Cojan, I., Mutz, S. G., Ehlers, T. A., and A., Mulch., Oxygen isotopic composition of Miocene meteoric water ($\delta^{18}\text{O}_w$) in Alpine foreland basins. (under review) *Geochemistry, Geophysics, Geosystems*.
- [2] Liu, X., Liu, Y., **Ballian, A.**, Bernecker, M., Fiebig, J., Rao, Y., Ding, L., and Mulch, A., 2026. Low Elevation and Humid Climate in the Eocene Gerze Basin, Central Tibet: Insights from Multi-Proxy Isotopic Analyses. *Global and Planetary Change*, 105547, ISSN 0921-8181. <https://doi.org/10.1016/j.gloplacha.2026.105547>.
- [3] Boateng, D., Mutz, S. G., Methner, K., **Ballian, A.**, Meijers, M. J. M., Mulch, A., and Ehlers, T. A., 2025. Refining paleoelevation estimates of the European Alps by simulating Middle Miocene climate and $\delta^{18}\text{O}$ responses to diachronous surface uplift scenarios. *Global and Planetary Change*, 251, 104808. <https://doi.org/10.1016/j.gloplacha.2025.104808>.
- [4] **Ballian, A.**, Meijers, M. J. M., Cojan, I., Huyghe, D., Bernecker, M., Methner, K., Tagliavento, M., Fiebig, J., and A., Mulch, 2025. Middle Miocene climate evolution in the Northern Mediterranean region (Digne-Valensole basin, SE France). *Climate of the Past*, 21(4), 841-856. <https://doi.org/10.5194/cp-21-841-2025>.
- [5] **Ballian, A.**, Chawchai, S., Miocic, J. M., Charoenchatree, W., Bissen, R., and Preusser, F., 2024. Late Holocene coastal dynamics south of the Chanthaburi estuary, eastern Gulf of Thailand. *Quaternary Research* 117, 19–29. <https://doi.org/10.1017/qua.2023.34>.
- [6] Bernecker, M., Hofmann, S., Davies, A., Tagliavento, M., Meijer, N., **Ballian, A.**, and Fiebig, J., 2023. A robust methodology for triple (Δ_{47} , Δ_{48} , Δ_{49}) clumped isotope analysis of carbonates. *Chemical Geology*, 642(July), 121803. <https://doi.org/10.1016/j.chemgeo.2023.121803>.
- [7] Boateng, D., Mutz, S. G., **Ballian, A.**, Meijers, M. J. M., Methner, K., Botsyun, S., Mulch, A., and Ehlers, T. A., 2023. The effects of diachronous surface uplift of the European Alps on regional climate and the oxygen isotopic composition of precipitation, *Earth Syst. Dynam.*, 14, 1183–1210. <https://doi.org/10.5194/esd-14-1183-2023>.